Practice workbook · Volume of revolution about the x-axis

Squaring y and finding pi times the integral of y^2 dx between limits, including solving for an unknown limit.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-22

1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

6 (a) Using a formula from page 2, show that $\cos 2A = 1 - 2\sin^2 A$

(2)

The finite region R is bounded by the curve with equation $y = 3 + 2\sin x$, the x-axis, the y-axis and the line with equation $x = \frac{\pi}{4}$

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.
Give your answer to the nearest integer.

(6)

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P 7 1 6 6 6 A 0 1 6 3 6

1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 8 marks)

1. Question 6 · part b · 6 marks · Working and answer

Working and answer

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2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

11 Given that

$$y = \frac{e^{4x}}{32}(8x^2 - 4x + 1)$$

(a) show that $\frac{dy}{dx} = x^2e^{4x}$

(5)

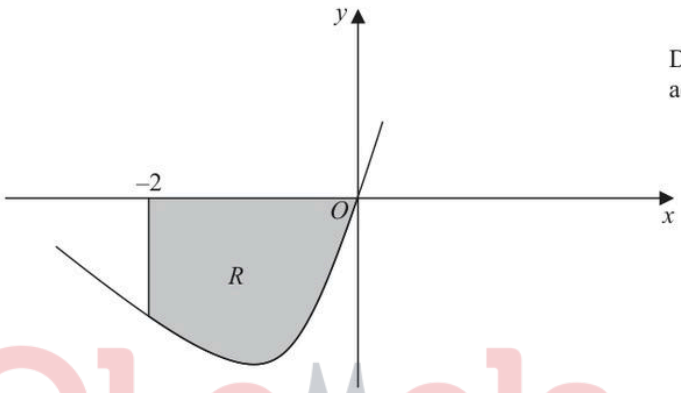


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C with equation $y = 3xe^{2x}$

The finite region R bounded by C, the straight line with equation $x = -2$ and the x-axis, shown shaded in Figure 2, is rotated though 360° about the x-axis.

(b) Using part (a), find the volume, to 2 significant figures, of the solid formed.

(4)

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2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 9 marks)

2. Question 11 · part b · 4 marks · Working and answer

Working and answer

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3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

11

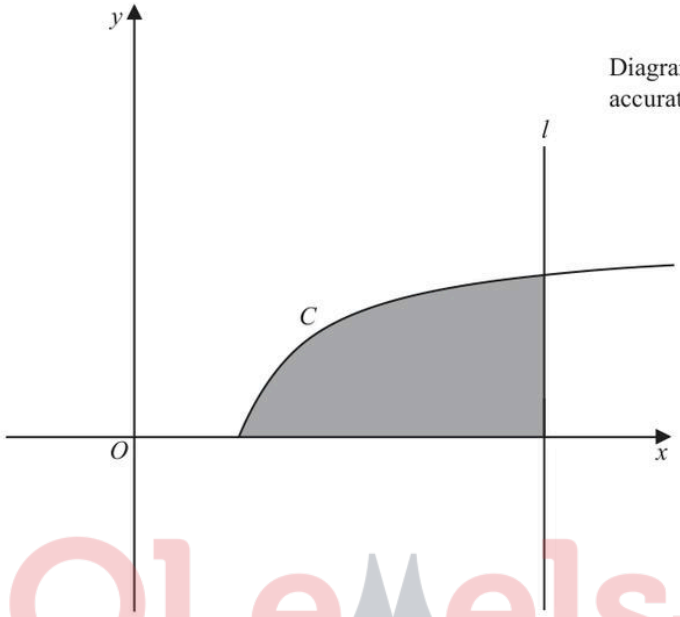


Figure 5

Figure 5 shows part of the curve C with equation $y = \sqrt{4x - 8}$ and the line l with equation $x = b$ where $b > 0$

The finite region bounded by the curve C, the x-axis and the line l, shown shaded in Figure 5, is rotated through $360°$ about the x-axis.

Given that the volume of the solid formed is 50π units^3

find the value of b

(7)

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3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

Question 11 continued

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3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

Question 11 continued

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3. Question 11 · part whole · 7 marks · Working and answer

Working and answer

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4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

2 The region enclosed by the curve with equation $y = e^{3x}$, the x-axis, the y-axis and the line with equation $x = 3$ is rotated through 360° about the x-axis.

Use algebraic integration to find, in terms of π and e, the volume of the solid generated.

(4)

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4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 4 marks)

4. Question 2 · part whole · 4 marks · Working and answer

Working and answer

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5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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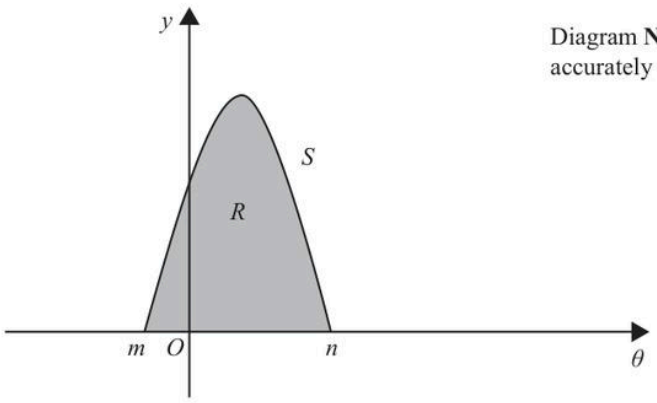


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve S with equation $y = \left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}}$

where $m \leqslant \theta \leqslant n$

The curve S meets the x-axis at the point with coordinates $(m, 0)$ and at the point with coordinates $(n, 0)$

(a) Find the exact value of m and the exact value of n

(3)

The finite region R, shown shaded in Figure 2, is bounded by the curve S, and the x-axis in the region $m \leqslant \theta \leqslant n$

The region R is rotated through 2π radians about the theta-axis.

(b) Use calculus to find the exact volume of the solid generated.

(4)

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5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 7 marks)

5. Question 7 · part b · 4 marks · Working and answer

Working and answer

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6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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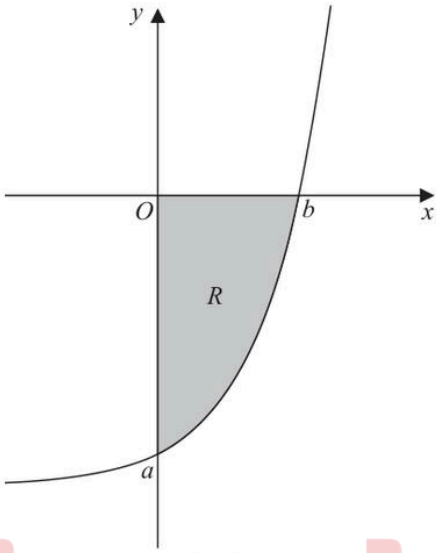


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Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where p and q are integers.

(6)

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6. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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6. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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6. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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6. Question 11 · part b · 6 marks

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Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 9 marks)

6. Question 11 · part b · 6 marks · Working and answer

Working and answer

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